MODULE 4 · RUNWAYS AND PERFORMANCE

Lesson 14: Performance Effects of Cold Weather and Contamination

Lesson 14 of 30 · about 8 min

LEARNING OBJECTIVES

- Explain how runway contaminants affect acceleration and stopping.
- Describe the performance penalties of anti-ice and contaminated runways.
- Recognise the operational restrictions on reduced thrust and V1.



Runway contaminants: standing water, slush and wet snow each have a different effect on performance.

Two families of effects

Aircraft contamination (Lesson 5) reduces lift and increases drag and stall speed. **Runway contamination** affects acceleration, stopping and directional control. This lesson covers the runway.

Acceleration

Loose, wet contaminants — standing water, slush, wet snow and deep dry snow — slow the aircraft down through:

- **displacement drag**: tyres pushing the contaminant aside;
- **impingement drag**: spray striking the airframe, gear and flaps.

The deeper and denser the contaminant, the greater the drag. Drag peaks close to the aquaplaning speed. The result is a longer take-off distance and, after an engine failure, a longer accelerate-go distance.

Stopping and control

- Braking friction is reduced on every contaminant. On ice it can be a fraction of a dry runway.
- Aquaplaning reduces wheel braking and steering almost to zero (Lesson 16).
- Crosswind limits are reduced, depending on the runway condition code.
- The **accelerate-stop distance** grows. To compensate, performance calculations usually produce a **lower V1** and may reduce the maximum take-off mass.

Operational rules

TYPICAL RESTRICTIONS — CHECK YOUR AFM

- **Reduced (assumed temperature / FLEX) thrust is not permitted on contaminated runways.** A fixed derate may be permitted when contaminated runway data exists for it.
- Use the contaminated runway performance data for the reported contaminant type and depth.
- Maximum contaminant depth for take-off is limited by the AFM — typically around 12.7 mm (½ in) of standing water, slush or wet snow, with higher limits for dry snow.

Anti-ice and cold temperatures

- Engine and wing anti-ice take bleed air, which reduces available thrust and climb performance. Performance tables include corrections when anti-ice is on.
- Cold, dense air normally *improves* engine and aircraft performance, but very low temperatures may bring limits on engine thrust, oil temperature or fluid use.
- Some types use higher approach or manoeuvring speeds when ice has accumulated on unprotected surfaces. Follow the FCOM's “ice accretion” speeds and performance corrections.

KEY TAKEAWAYS

- Wet contaminants add drag and lengthen take-off; all contaminants reduce braking.
- Contaminated runways usually mean a lower V1 and possibly a lower maximum take-off mass.
- No assumed-temperature (FLEX) reduced thrust on contaminated runways.
- Anti-ice reduces thrust — apply the performance corrections.

KNOWLEDGE CHECK

What is the usual rule on assumed temperature (FLEX) reduced thrust when the runway is contaminated?

A. It is always required to protect the engines

B. It is not permitted

C. It is permitted if V1 is reduced by 10 kt

D. It is permitted at night only

Answer: B. Assumed temperature reduced thrust is not permitted on contaminated runways. A fixed derate may be allowed if the AFM provides contaminated runway data for it.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.